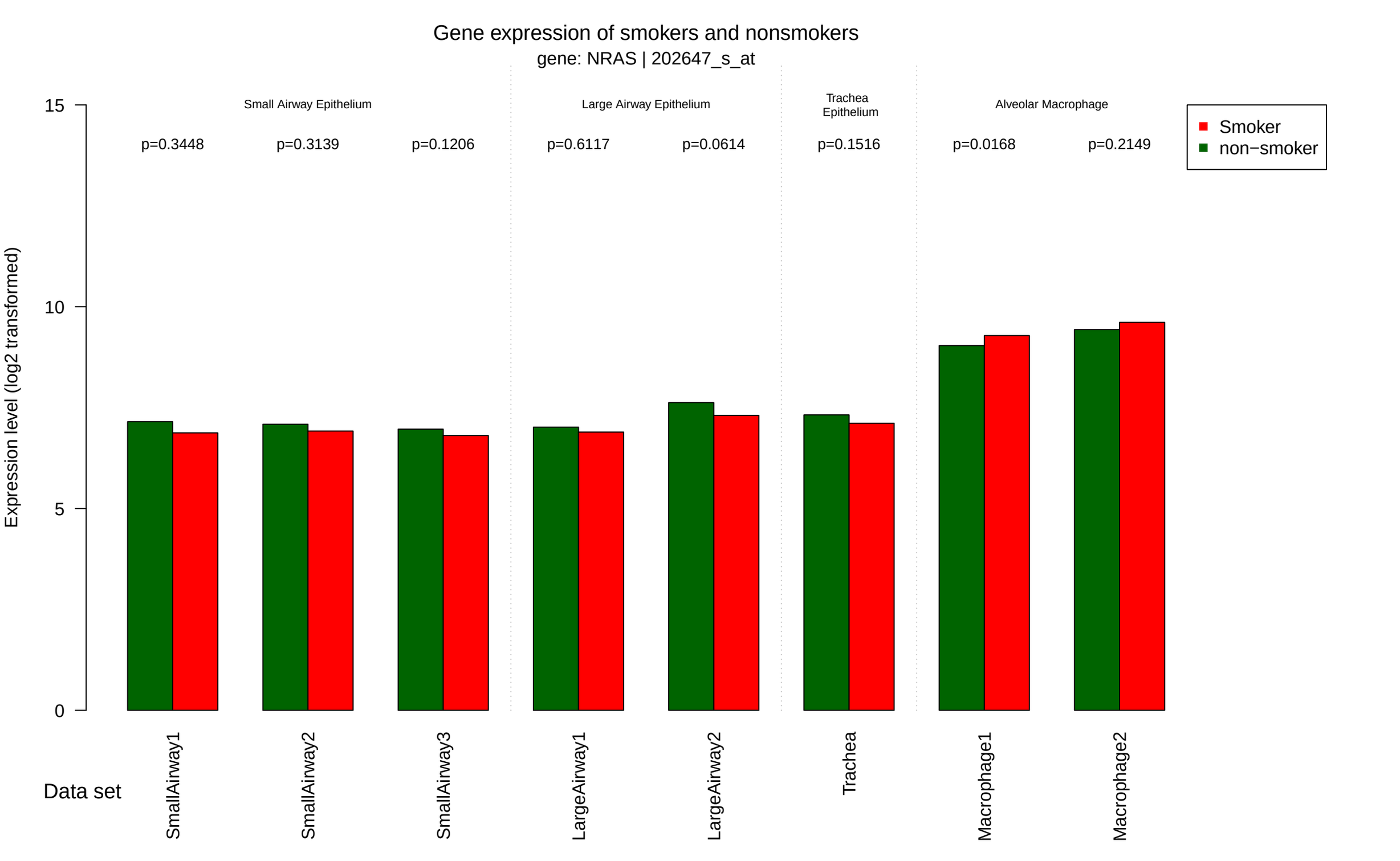
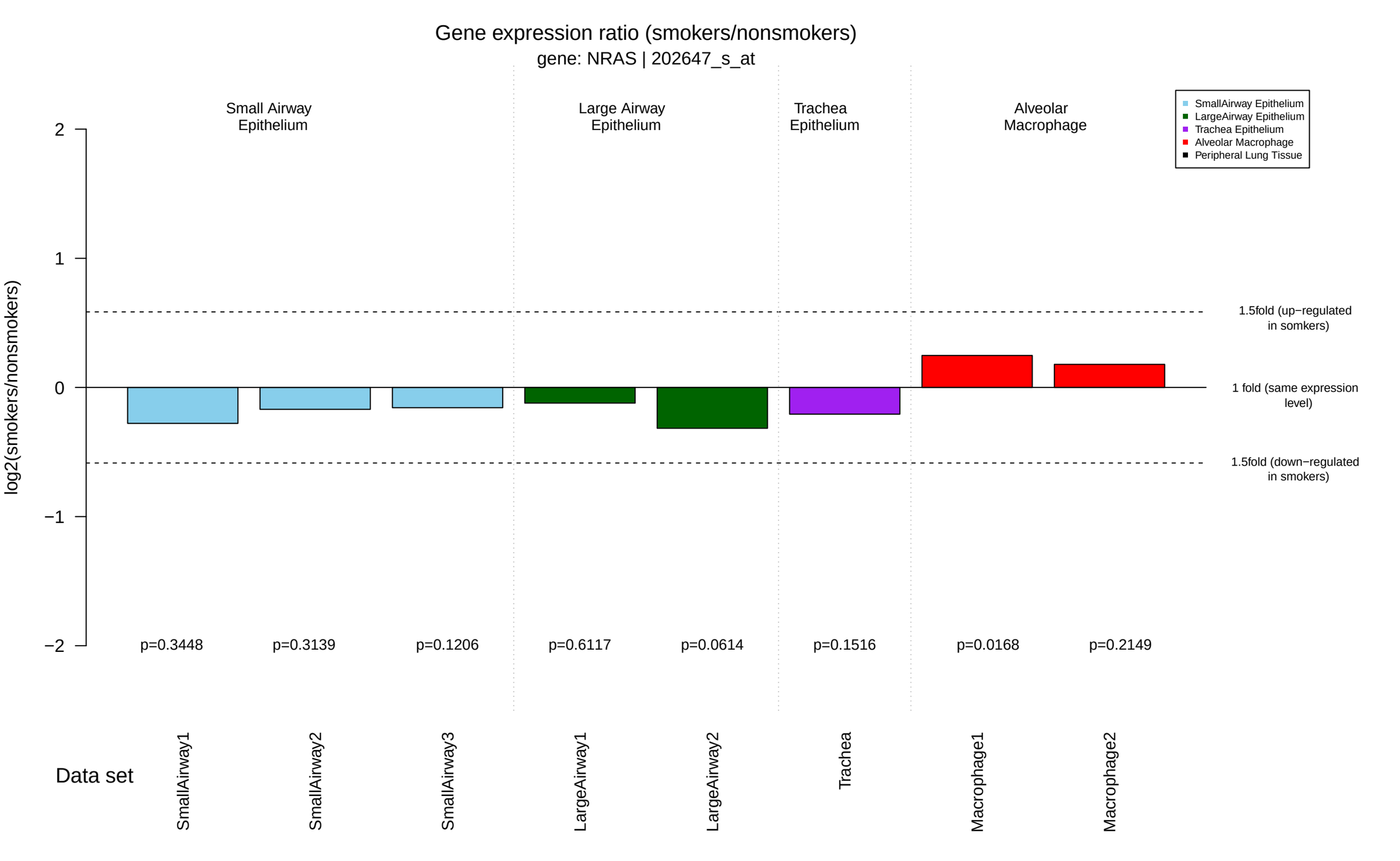
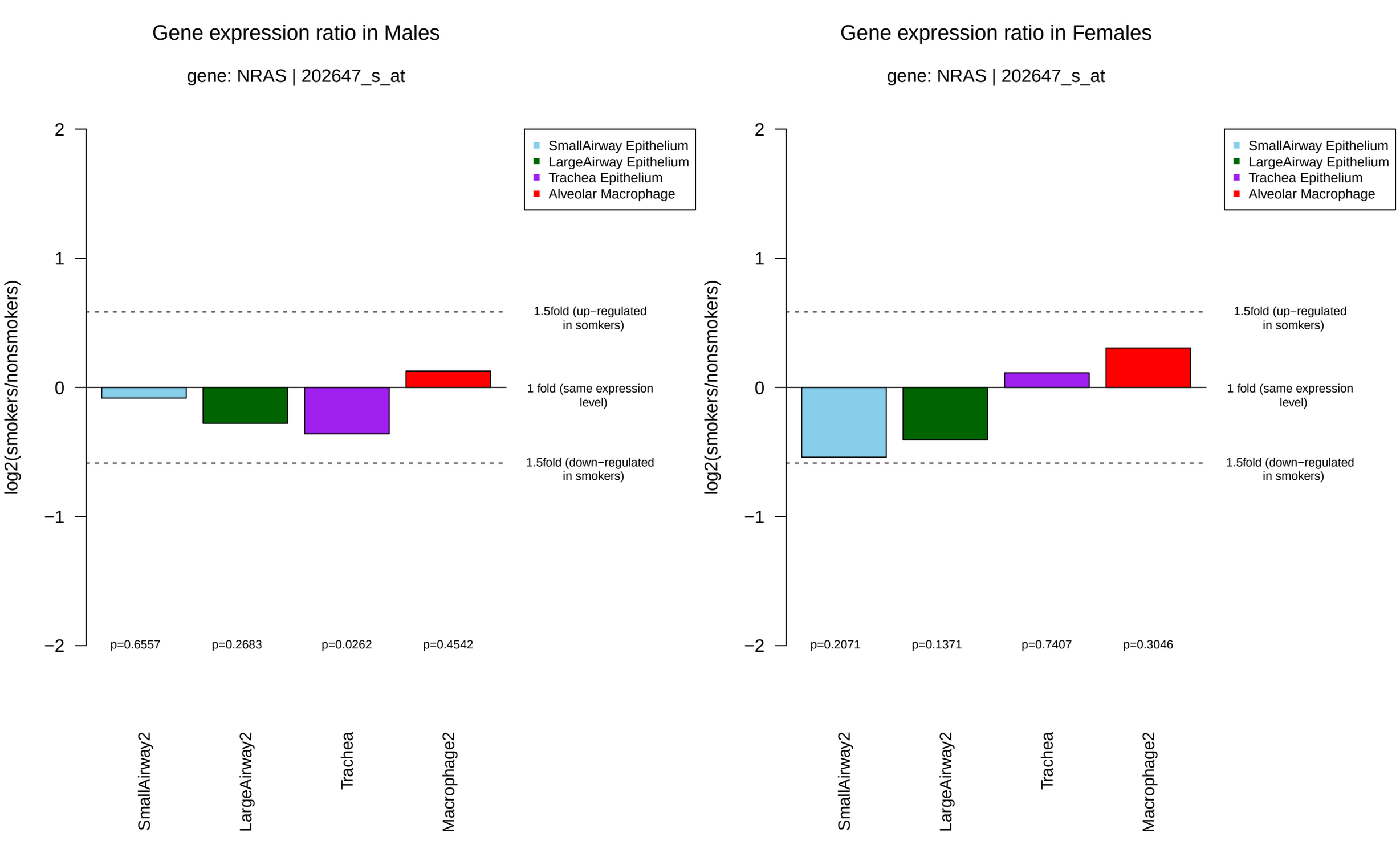
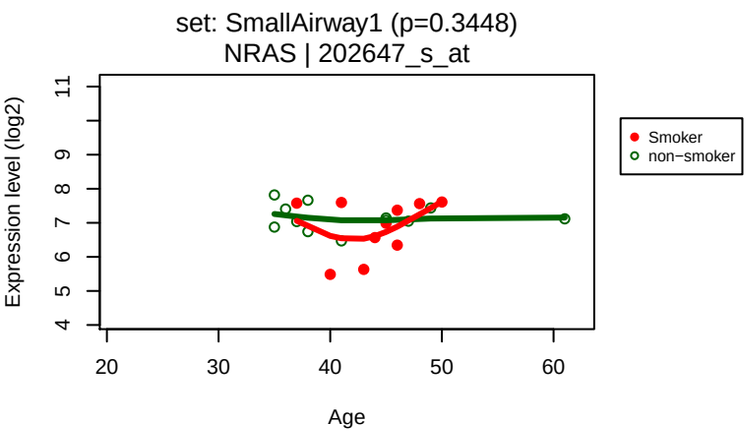
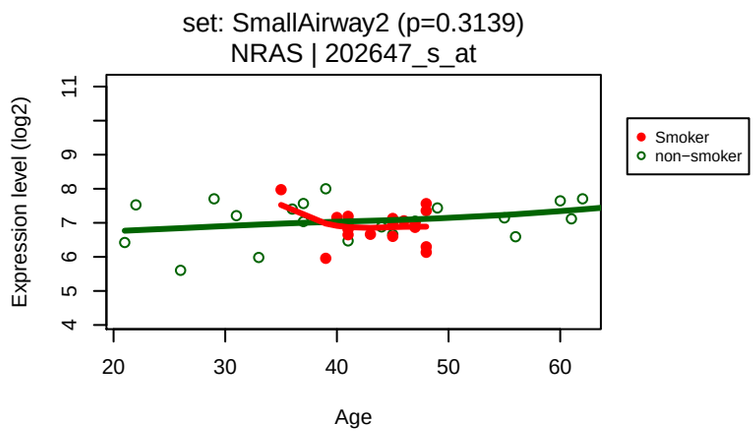
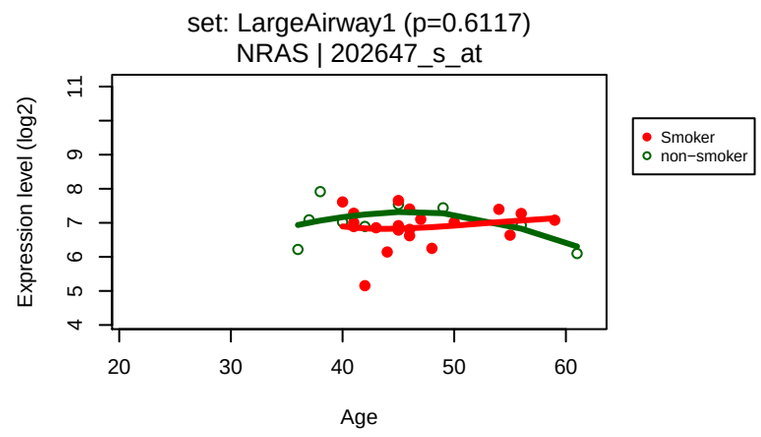

Gene expression of smokers and nonsmokers
gene: NRAS | 202647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3448
p=0.3139
p=0.1206
p=0.6117
p=0.0614
p=0.1516
p=0.0168
p=0.2149
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

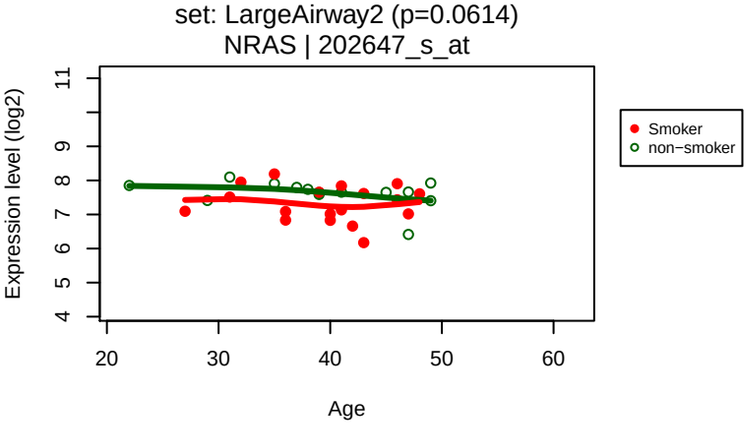

Gene expression ratio (smokers/nonsmokers)
gene: NRAS | 202647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3448
p=0.3139
p=0.1206
p=0.6117
p=0.0614
p=0.1516
p=0.0168
p=0.2149
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

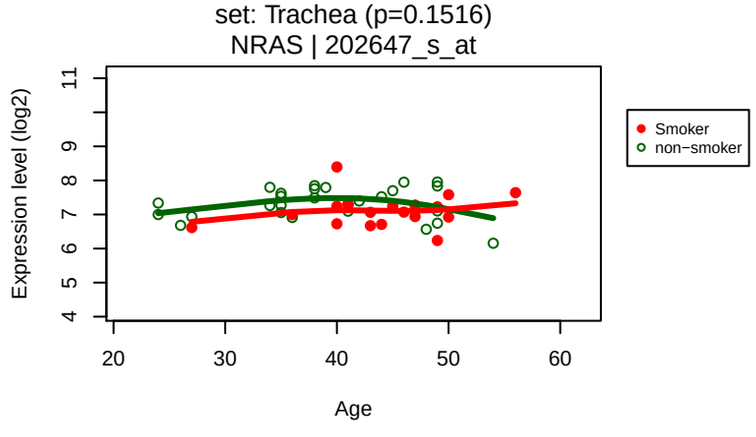

Gene expression ratio in Males
gene: NRAS | 202647_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6557
p=0.2683
p=0.0262
p=0.4542
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NRAS | 202647_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2071
p=0.1371
p=0.7407
p=0.3046
SmallAirway2
LargeAirway2
Trachea
Macrophage2

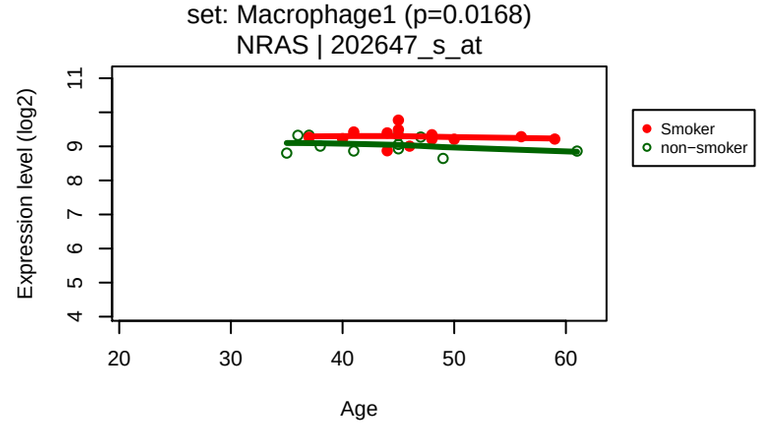

set: SmallAirway1 (p=0.3448)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker

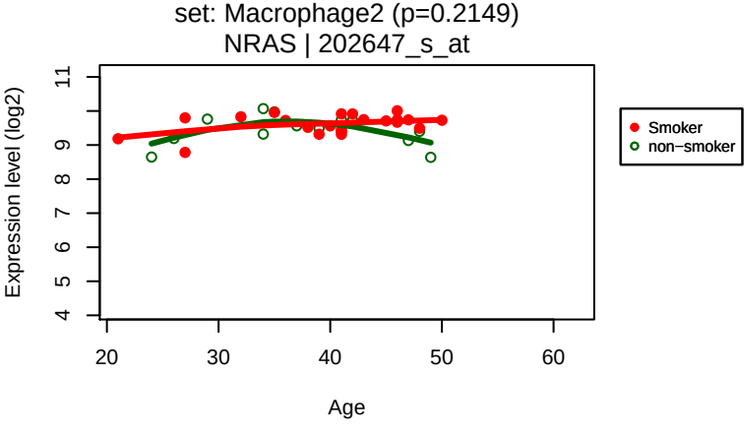

set: SmallAirway2 (p=0.3139)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker

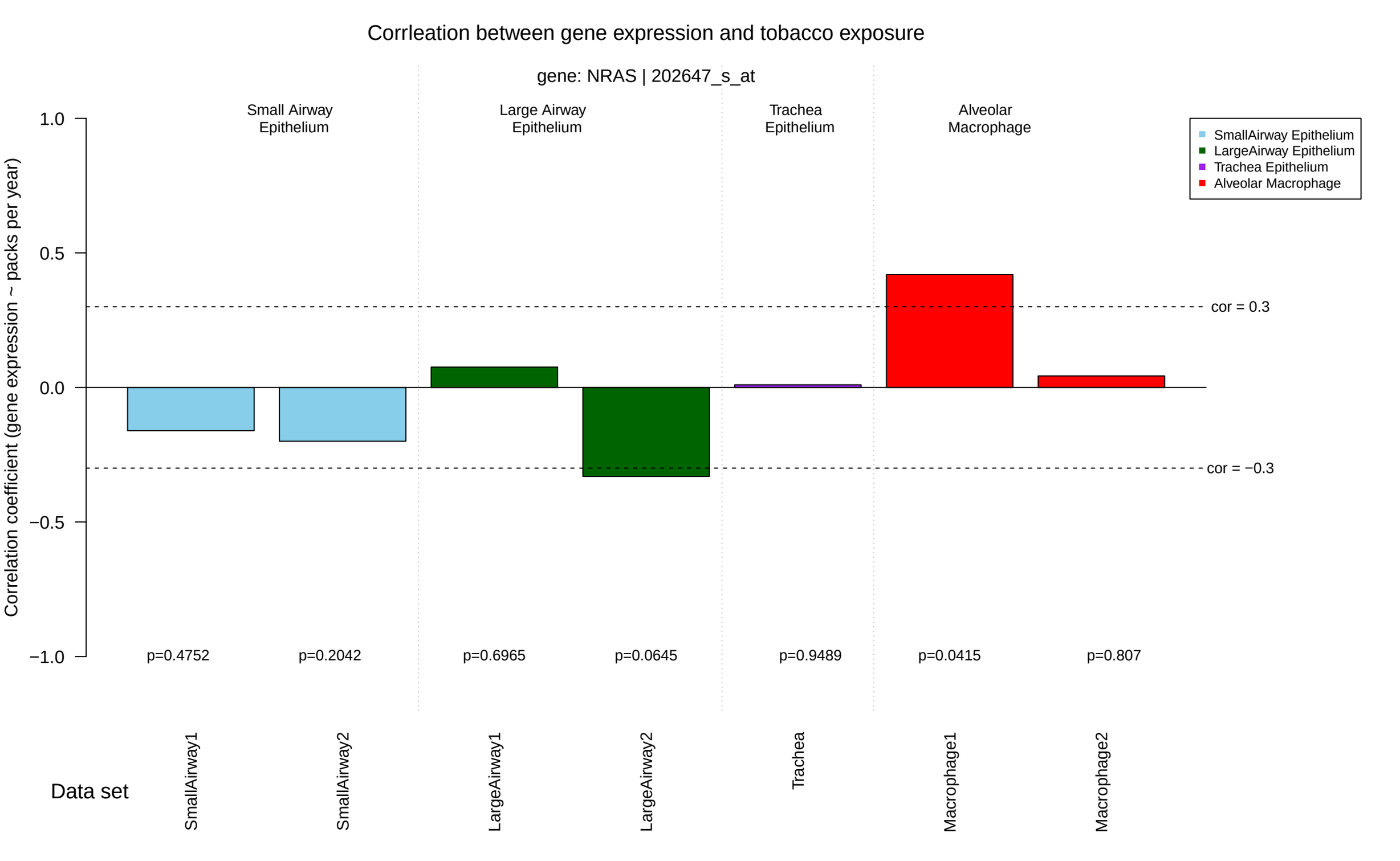

set: LargeAirway1 (p=0.6117)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0614)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1516)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0168)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2149)
NRAS | 202647_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NRAS | 202647_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4752
p=0.2042
p=0.6965
p=0.0645
p=0.9489
p=0.0415
p=0.807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2